

Project Initialization and Planning Phase

Date	15 July 2026
Team ID	
Project Title	SkillEcho - Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform oral concept assessment using AI, boosting objectivity and scale. It tackles subjective, unscalable manual grading, promising better evaluation consistency, full audit trails, and happier evaluators and students. Key features include an ASR + NLP + DSP evaluation pipeline and a deterministic 0-100 competency score.

Project Overview

Field	Details
Objective	Automate the evaluation of spoken conceptual explanations by combining Speech-to-Text, semantic NLP, and acoustic DSP to produce an objective, reproducible 0-100 competency score.
Scope	Covers university viva/oral-exam and HR technical-interview assessment: account management, concept library, audio submission, AI evaluation pipeline, PDF reporting, and admin oversight.

Problem Statement

Field	Details
Description	Manual oral assessment is subjective (evaluators score the same answer differently), lacks acoustic metrics of delivery, is time-consuming (8-12 students/hour), and leaves no standardised audit trail for grade disputes.
Impact	Inconsistent grading and evaluator fatigue reduce assessment quality and scalability; solving this yields objective, reproducible scores, a ~90% reduction in per-student evaluation time, and a permanent digital audit trail.

Proposed Solution

Field	Details
Approach	Combine Whisper ASR transcription, dual-model semantic scoring (Bi-Encoder + Cross-Encoder), Librosa-based acoustic/filler analysis, and a deterministic weighted scoring matrix (0-100) delivered through a Streamlit + FastAPI cloud application.
Key Features	- Whisper Speech-to-Text transcription - Bi-Encoder + Cross-Encoder semantic similarity scoring - Acoustic feature extraction (RMS, pause ratio, ZCR) and filler-word analysis - Topic Guardrail to flag off-topic responses before heavy inference - Deterministic 0-100 weighted scoring engine with 3-tier classification - Downloadable ReportLab PDF evaluation report - Admin concept management (CRUD with PDF attachments) and evaluation history

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources - CPU specification, no GPU	1 vCPU (Streamlit Community Cloud, CPU-only instance)

Resource Type	Description	Specification / Allocation
	Memory - RAM specification	< 1 GB RAM (hard platform constraint)
	Storage - Disk space for data, models, and logs	Ephemeral local disk for uploads; persistent Neon PostgreSQL for data
Software	Frameworks - Python frameworks	FastAPI, Streamlit
	Libraries - Additional libraries	openai-whisper, librosa, sentence-transformers, reportlab, sqlalchemy, psycpg2-binary
	Development Environment - IDE	VS Code, Python 3.11 virtual environment (venv311)
Data	Source, size, format	5 seeded reference concepts (ML, DL, NLP, RL, Neural Networks) - text-based concept library, no external dataset required